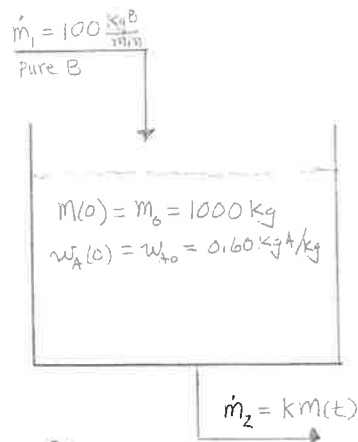


To Receive Full Credit, You Must Show All Your Work

Initially at $t = 0$, a mixing tank contains $m(0) = 1000$ kg of a binary liquid mixture of species A and B. The initial mass fraction of species A is $w_A(0) = 0.60$. The inlet flow rate is $\dot{m}_1 = 100$ kgB/min of pure B. The outlet flow rate is given by $\dot{m}_2 = k * m(t)$ where $k = 2.0 \text{ min}^{-1}$ and $m(t)$ is the mass in the tank as a function of time. Assume the tank is well mixed throughout the process.



- (a) At steady state, how much mass, in kg, is left in the tank? That is, when $t \rightarrow \infty$, what is $m(t \rightarrow \infty)$?
- (b) Derive a general expression for $m(t)$. How long does it take for $m(t)$ to drop to 100 kg?
- (c) Derive a general expression for $m_A(t)$, the mass of A in the tank. At the time calculated in part (b), what is w_A , the mass fraction of A in the tank?

$$\text{Accum} = \text{in} - \text{out} + \text{gen} - \text{con}$$

$$t = 1 \text{ min}$$

$$\dot{m}_2 = k m(t)$$

$$m(t) = \frac{\dot{m}_2}{k}$$

a)

$$m(0) = 1000 \text{ kg}$$

$$\dot{m}(t) = \frac{2}{\dot{m}_2}$$

$$\text{Accum} = \dot{m}_1 - \dot{m}_2 + m(t)$$

$$0 = \dot{m}_1 - \dot{m}_2$$

$$\frac{dm}{dt} = \dot{m}_1 - \dot{m}_2$$

$$\frac{dm}{dt} = 100 \frac{\text{kg}}{\text{min}} - 2m(t)$$

$$\int \frac{dm}{100 - 2m(t)} = \int dt$$

$$\ln(100 - 2m(t)) = -2t$$

$$m(t) = 100 e^{-2t}$$

$$m(t) = 100 e^{-2t}$$

$$m(t) = 100 - 100$$

$$m(t) = 900 \text{ kg}$$

$$\text{Accum}_A = \dot{m}_1 - \text{out} = -k m(t) (w_A(t)) + 1000 \text{ kg}$$

$$\text{Accum}_B = 100 \text{ kgB/min} - (1 - w_A(t)) k m(t)$$

$$\frac{dm_A}{dt} = 1000 - 2m(t) w_A(t) = \int \frac{dm_{A \text{ out}}}{1000 - 2m_{A \text{ out}}} = \int dt \Rightarrow \ln(1000 - 2m_{A \text{ out}}) = t$$

$$\frac{dm_B}{dt} = 100 - (1 - w_A(t)) 2m(t)$$

$$\Rightarrow \frac{dm_B}{dt} = 100t - m_{B \text{ out}}$$

